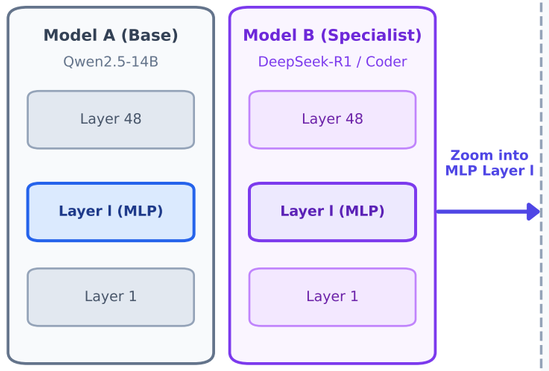
How Phylogenetic Tree Surgery Works on Transformer LLMs (Concept & Method)

1. Treat MLP Neurons as Homologous Species → 2. Build Layer Trees of Clades (Sub-Circuits) → 3. Detect Partner Swaps (Quartet Violations) → 4. Perform Matrix Surgery

STEP 1 • THE MODELS — Two Transformer Models & Inside the MLP Block

Each MLP layer has 13,824 neurons — same index i across Base and Specialist models

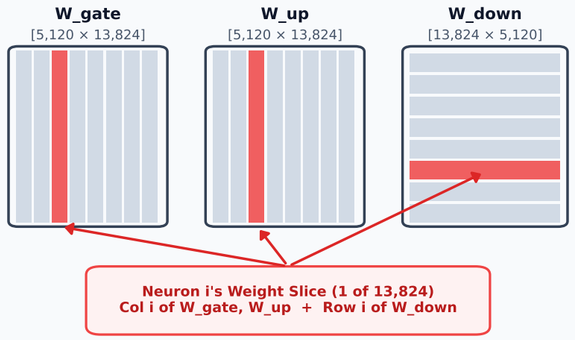
Two Sibling 14B Transformer Models (Shared Ancestor Initialization)



Zoom into MLP Layer l

Inside One Transformer SwiGLU MLP Layer (13,824 Neurons)

$MLP(x) = \Sigma_{i=1..13,824} [SiLU(w_gate,i \cdot x) \times (w_up,i \cdot x)] \cdot w_down,i$

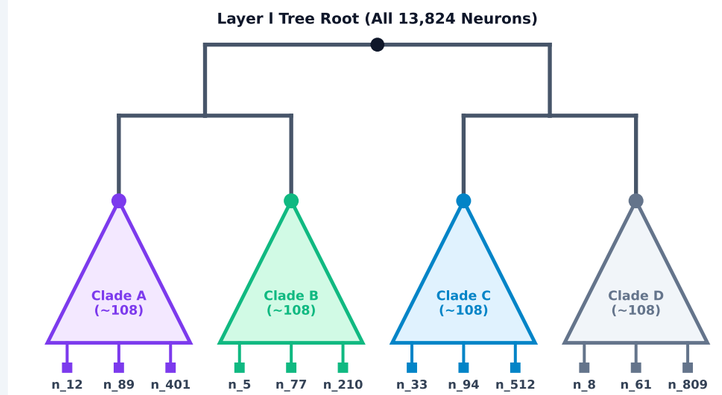


What is a 'Species' here? Each MLP neuron i (out of 13,824 in layer l) is defined by its input/output weight vectors. Because Model A and Model B started from the same checkpoint, Neuron #412 in Model A is the SAME ancestral neuron as Neuron #412 in Model B!

STEP 2 • THE TREE — What the Tree Is: Leaves = Neurons, Clades = Circuits

Clustering neurons by weight similarity partitions the MLP matrix into functional clades

We Group the 13,824 Neurons in Layer l into a Binary Phylogenetic Tree by Weight Similarity

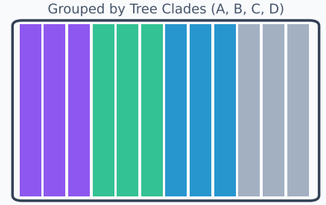


Leaves = Individual MLP Neurons (1..13,824) | Clades = Subtrees of Co-Adapted Neurons

Maps to MLP Slices

What a Clade Means in ML Terms

Each Clade = A Multi-Neuron Sub-Circuit!
MLP Weight Matrix W_layer
Grouped by Tree Clades (A, B, C, D)



- Single neurons are too small / polysemantic
- Whole layers (13,824 neurons) are too coarse
- A Tree Clade (~100 neurons) is the natural functional building block (organelle) of an MLP!

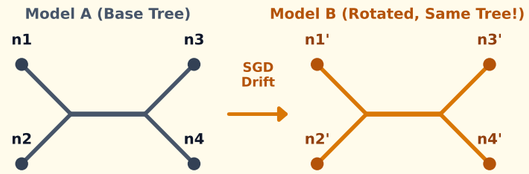
What is the Tree? Just as a phylogenetic tree in biology groups species into families & genera, our GPU BME tree groups the 13,824 neurons in each Transformer layer into ~128 evolutionary clades (6,171 clades across all 48 layers).

STEP 3 • REWIRING — What Tree Changes Mean: Drift vs. Partner Swaps

Why Tree Quartet Violations isolate real circuit rewiring while ignoring uniform SGD drift

Case A: Uniform Weight Drift / Rotation

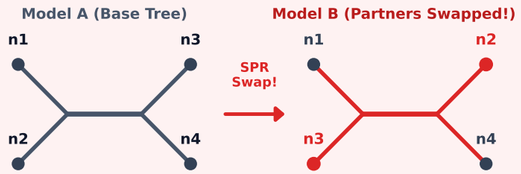
Why Standard Weight Diffs (||W_B − W_A||) Get Fooled



- Every weight vector moved (||W_B − W_A|| is LARGE)
- Magnitude / TIES / Cosine think this clade changed!
- But tree partners {n1, n2} | {n3, n4} are IDENTICAL
→ Our Tree Quartet Score = 0 (Ignored as harmless drift)

Case B: Tree Quartet Violation (Partner Swap)

True Functional Circuit Rewiring Between Models



- **Neurons n2 and n3 SWAPPED evolutionary partners:**
Base: {n1, n2} | {n3, n4} → Specialist: {n1, n3} | {n2, n4}
- **Who fires with whom inside the MLP block changed!**
→ High Quartet Violation Score (Flagged for Tree Surgery)

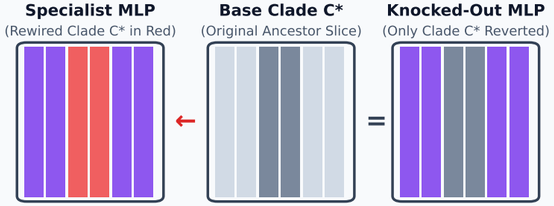
The Core Idea: Instead of asking 'Which individual weights moved the most?' (which picks up noisy SGD drift), we ask a relational question: 'Which tree clades broke their neighborhood topology (swapped quartet partners) during fine-tuning?'

STEP 4 • SURGERY — How Tree Surgery Happens on Transformer Matrices

Using tree clades as a surgical index mask for Knock-Out (ablation) and Chimera (grafting)

Surgery Mode 1: Clade Knock-Out (Ablation)

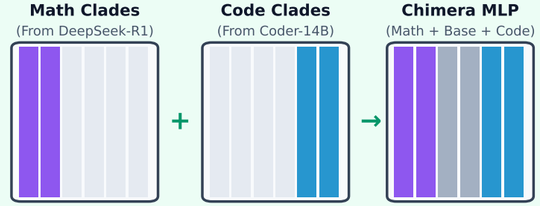
Test If a Rewired Clade Is Causally Necessary for a Skill



- How it works: Keep 99% of the Reasoning Model intact, but overwrite ONLY the columns/rows of Clade C* in W_gate, W_up, W_down with the Base Model's weights.
- What it means: If reasoning collapses while general text stays intact, Clade C* is the actual causal reasoning circuit!

Surgery Mode 2: Clade Grafting (Chimera)

Combine Specialist Circuits into One Model (Zero Training)



- Why 50/50 weight averaging fails: Averaging two models blends every neuron half-and-half, scrambling non-linear circuits.
- How Tree Grafting works: Transplant whole, intact Math clades (purple slices) and non-overlapping Code clades (blue slices) into the Base MLP block — zero training or fine-tuning!

Why 'Tree Surgery' Works: Because SwiGLU MLPs are a sum over 13,824 neurons, replacing or grafting an entire tree clade C (cols C of W_gate, W_up and rows C of W_down) swaps a complete, self-contained functional sub-circuit without breaking the layer!